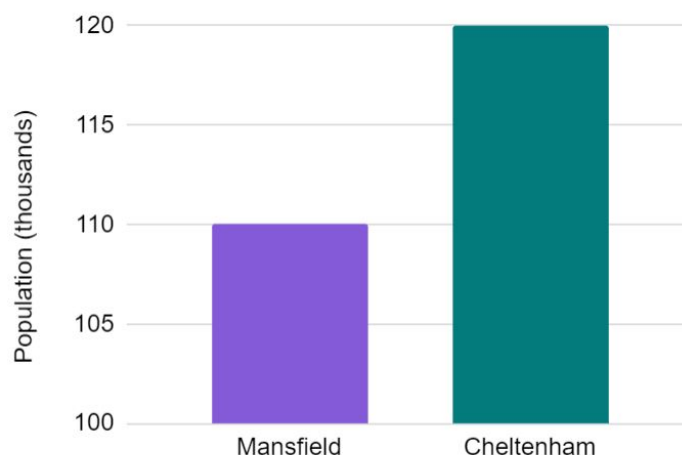
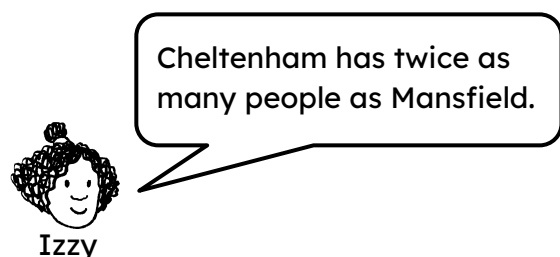




Statistical problems - data presentations

Task A: Presenting data clearly using graphs

1) The graph shows data from the ONS about the population of two towns in 2019.



a) Explain why Izzy might think this, based on the bar chart.

The bar for Cheltenham is twice as long as the bar for Mansfield.

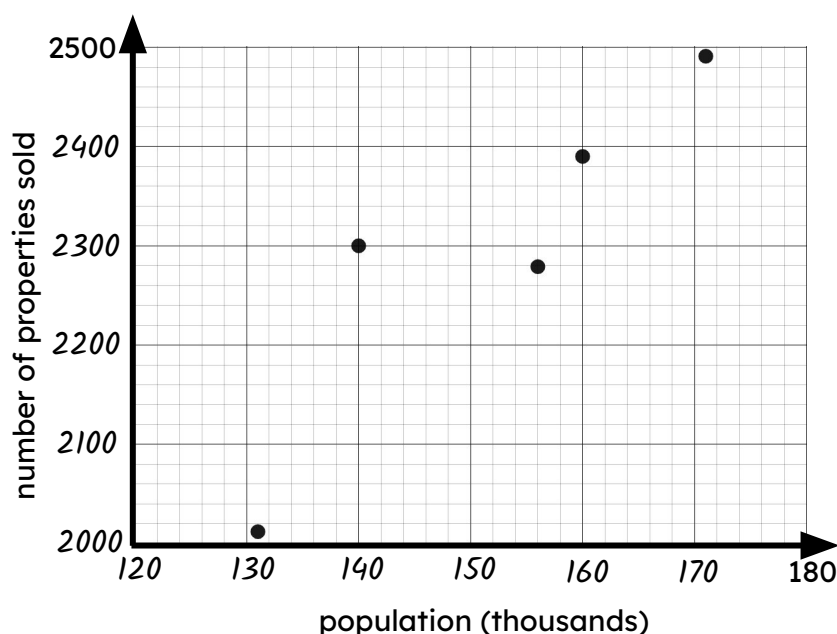
b) Explain why this is wrong.

The scale does not start at zero. There are 110 thousand people in Mansfield and 120 thousand people in Cheltenham.

2) The table shows data from the ONS about five towns in 2019.

Complete the scale on the axes of the scatter graph.

population (thousands)	number of properties sold
131	2012
140	2300
156	2279
160	2390
171	2491

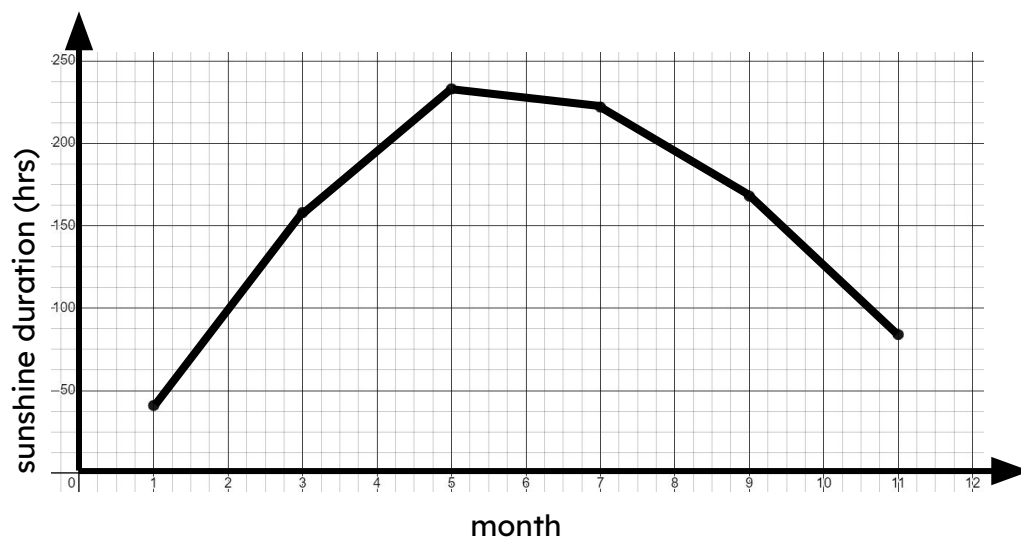


Task B: Choosing how to present data to illustrate a point

1) The table shows weather data from The Met Office for Cardiff in 1995.

month	sunshine duration (hours)
1	41
3	158
5	233
7	222
9	168
11	84

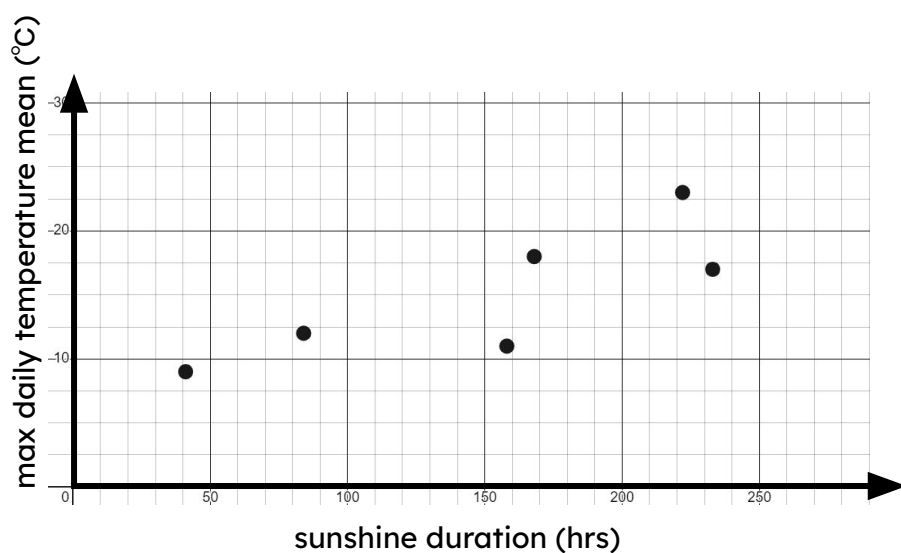
Plot a graph to illustrate how the duration of sunshine changed during the year.



2) The table shows weather data from The Met Office for Cardiff in 1995.

sunshine duration (hours)	max daily temperature mean ($^{\circ}\text{C}$)
41	9
158	11
233	17
222	23
168	18
84	12

Plot a graph to illustrate the relationship between sunshine duration and maximum temperature.



3) The table shows weather data from The Met Office for four locations in 1995.

location	days with airfrost
Ballypatrick Forest	23
Cardiff	33
Durham	64
Lerwick	59

Plot a graph to illustrate how the total number of days with airfrost differs between the four locations.

